



Department of Mechanical Engineering

Dear NSF GRFP Selection Committee,

I am writing to enthusiastically recommend Mr. Jacob Vaught for the NSF Graduate Research Fellowship. I have supervised Jacob's work in our radar engineering research group for the past two years, during which he has emerged as an extraordinarily capable and innovative young engineer. I have worked with many students in my career, and Jacob stands out in the top 5% for his technical aptitude, initiative, and maturity. It is no exaggeration to say that Jacob operates at a graduate level of research already – he is one of the most impressive researchers I have encountered in radar engineering.

As a research faculty member specializing in radar systems, I had the pleasure of overseeing Jacob's contributions to several advanced radar projects. Jacob's intellectual merit is demonstrated by the deep technical contributions he has made to our lab's high-frequency radar platform. In particular, he spearheaded multiple innovative engineering initiatives that significantly improved our radar system's performance.

Jacob developed a customizable control interface for our radar, allowing dynamic reconfiguration of waveform parameters and processing settings. This software extension enabled experiments and operating modes beyond the stock capabilities of the equipment, showcasing his creativity in software radar design.

He also implemented novel multi-gain and multi-pulse radar imaging techniques to achieve high dynamic range detection. By cleverly varying gain settings and interleaving pulses, Jacob's approach improved the radar's sensitivity to weak targets by over 6 dB. This enhancement means our system can now detect small, distant objects (like small boats) that previously fell below the noise floor – a remarkable achievement.

To accelerate our research, Jacob built a comprehensive simulation package that emulates our radar's signal processing pipeline. This tool allows us to test detection algorithms on recorded and synthetic data before deploying to hardware, greatly speeding up development. Jacob's simulator has become a core tool, and its reliability reflects his rigorous engineering standards.

These technical accomplishments would be commendable for an experienced graduate student, and Jacob achieved them with minimal supervision. For example, when we encountered unexplained noise in field data, Jacob took the initiative to debug the issue. He traced it to a timing mismatch in the multi-pulse schedule and then implemented a firmware patch in the control software to fix it. I vividly recall arriving at the lab one morning to find Jacob had spent the night resolving a problem in our camera system –

by sunrise, it was running well enough to be used for a data capture. This kind of determination and skill in troubleshooting is exceptional. Jacob's hands-on problem-solving style and persistence through challenges are exactly what one hopes for in a researcher. He not only identifies problems but attacks them methodically until resolved, demonstrating resilience and a deep understanding of the system.

Beyond his raw technical talent, Jacob shows strong potential for broader impacts in science and engineering. He is eager to share his knowledge and uplift those around him. In our lab, Jacob has taken on a mentorship role with younger students – he routinely trains new members on radar operation, signal processing techniques, and safety protocols. It is common to see Jacob in front of a radar screen with two newer students at his side, patiently explaining how to interpret spectra or helping them troubleshoot their code. This willingness to teach others and build our lab's collective expertise is a clear indicator of his collaborative spirit and leadership. Jacob's enthusiasm is infectious, and he has a natural ability to communicate complex ideas in an accessible way – a trait that will serve the STEM community well.

I am particularly impressed by how Jacob has integrated his radar work with broader, interdisciplinary goals. His proposed graduate research involves fusing radar and camera data to detect small maritime targets, a project that sits at the intersection of hardware engineering and machine learning. Jacob has shown great foresight by already laying groundwork for this endeavor. He has been collaborating with our lab's camera and sensor team to align radar data with visual imagery. In doing so, he's effectively bridging the gap between traditional radar engineering and modern computer vision. This ability to cross domains underscores Jacob's versatility and creativity – he not only masters the details of his own field, but also reaches out to integrate new techniques (for example, he learned fundamentals of computer vision and even some machine learning to better combine the data).

The Intellectual Merit of Jacob's work is evident in this interdisciplinary approach; he is advancing knowledge by creating a novel radar-camera fusion methodology for detecting small vessels, which is an unsolved challenge crucial for coastal surveillance and search-and-rescue operations. Equally important are the Broader Impacts. Jacob's research has clear real-world relevance for maritime safety, and he plans to share the tools and data developed (he has already started curating a unique dataset of radar data of small boats, which he intends to release to the community). This speaks to Jacob's understanding that science should benefit society at large.

As someone who has worked closely with him on challenging engineering problems, I can attest that Jacob meets and exceeds the NSF's criteria for Intellectual Merit and Broader Impacts. He is ready to be a leader in our field. I give him my strongest recommendation for the NSF GRFP – Jacob has my full confidence and highest endorsement for this fellowship.

Sincerely,
Dr. Douglas Cahl
Research Assistant Professor
University of South Carolina, Dept. of Mechanical Engineering